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Paleomagnetism and geochronology of Early Cretaceous volcanic rocks in the eastern segment of the Lhasa terrane, Tibetan Plateau, and their tectonic implications

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ABSTRACT

Paleogeographic reconstructions of the Lhasa terrane for the Cretaceous provide important constraints on the evolution of the Neo-Tethys Ocean and crustal shortening within Asia. However, the Cretaceous paleogeography of the Lhasa terrane remains contentious. A direct way to study this issue is to conduct paleomagnetic investigations of the Cretaceous rocks of the Lhasa terrane; however, most previous Cretaceous paleomagnetic investigations of the Lhasa terrane were conducted in the middle and western segments of this terrane. Different vertical-axis rotations affected different parts of the Lhasa terrane following the India-Asia collision, and therefore paleomagnetic data from the western and middle segments of the Lhasa terrane cannot necessarily be used to directly constrain the paleolatitudes of its eastern segment. This study presents paleomagnetic data from the Early Cretaceous volcanic rocks with well-constrained ages from the Luolong area in the eastern segment of the Lhasa terrane. Isotope geochronology reveals that these rocks formed at 127–124 Ma. The mean high-temperature direction obtained from 17 paleomagnetic sites is declination/inclination (*Ds*/*Is*) = 21.8°/19.0° with $\kappa s = 65.8$ and $\alpha_{95} = 4.4°$ (κ—best estimate of the precision parameter; α_{95}—radius of the 95% probability ellipse around the mean direction; s—stratigraphic coordinates). Petrographic investigations, a positive fold test, a reversal test, and a paleosecular variation test indicate the primary origin of this characteristic remanence. A paleomagnetic pole of 60.9°N, 227.2°E with *dp*/*dm* = 2.4°/4.6° (dp/dm—semi-axes of the 95% probability ellipse around the mean pole) yields a paleolatitude of 9.2 ± 2.4°N for the eastern segment of the Lhasa terrane. Combined with reliable results from previous paleomagnetic studies, we draw the following conclusions. (1) During the Early Cretaceous, the Lhasa terrane was oriented WNW-ESE as a whole, and the eastern-middle segments may have been oriented nearly E-W. (2) Asia has accommodated 2050 ± 230 km of N-S crustal shortening along 96°E longitude since the Early Cretaceous. (3) The minimum N-S width of the Neo-Tethys Ocean at ca. 125 Ma was 4185 ± 300 km.

INTRODUCTION

The formation and evolution of the Tibetan Plateau reflect the dynamics of the Asian continent, and they have substantially influenced the paleoclimate of Asia (e.g., Tapponnier et al., 2001; Xu et al., 2006; Royden et al., 2008; Molnar et al., 2010; Ding et al., 2014; Yuan et al., 2021; Zhu et al., 2022). The plateau was formed by the collision and amalgamation of multiple tectono-stratigraphic terranes. This process was accompanied by the birth, growth, and decline of the Paleo-Tethys, Bangong-Nujiang Tethys, and Neo-Tethys Oceans. The evolution of the Neo-Tethys Ocean ended in the collision between India and Asia, which has been the main driver of the uplift of the Tibetan Plateau during the Cenozoic (e.g., Harrison et al., 1992; Coleman and Hodges, 1995; Yin and Harrison, 2000; Wu et al., 2008; Wang et al., 2014). Consequently, it is important to quantify the history of the Neo-Tethys Ocean to constrain the tectonic evolution of the Tibetan Plateau. In addition, the closure of the Bangong-Nujiang Tethys Ocean during the Late Jurassic through to the Early Cretaceous (e.g., Kapp et al., 2007; Yan et al., 2016; Zhu et al., 2016; Fan et al., 2018; Cao et al., 2020; Hu et al., 2022) resulted in significant intracontinental crustal shortening. This shortening may have led to the central Tibetan Plateau being raised to an elevation of 3–4 km prior to the India-Asia collision (e.g., Murphy et al., 1997; Kapp et al., 2005, 2007; Volkmer et al., 2007; Zhang et al., 2012; Ding et al., 2014; Xu et al., 2016). Thus, quantification of crustal shortening within Asia after the closure of the Bangong-Nujiang Tethys Ocean is also important for constraining the evolution of the Tibetan Plateau.

Despite decades of studies of the crustal shortening in Asia and the evolution of the Neo-Tethys Ocean, certain issues are still debated. For example, estimates of the width of the Neo-Tethys Ocean during the Early Cretaceous vary from ~4750 km (van Hinsbergen et al., 2011) to ~7500 km (Chen et al., 2012; Yang et al., 2015a; Ma et al., 2018; Li et al., 2022). Previous studies have proposed that the India-Asia collision has resulted in ~1400 km of crustal shortening, of which ~600 km have

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been accommodated in Asia (Yin and Harrison, 2000; Johnson, 2002; Volkmer et al., 2007). A synthesis of geological evidence indicates that since ca. 50 Ma, ~600 and ~1050 km of shortening have occurred within eastern and western Tibet, respectively (van Hinsbergen et al., 2011). Existing paleomagnetic data indicate that several hundred kilometers (Tan et al., 2010; Chen et al., 2017; Meng et al., 2018) to ~1500 km (Sun et al., 2012; Cao et al., 2017a; Li et al., 2022) of crustal shortening occurred within Asia since the Cretaceous.

The Cretaceous paleogeography of the Lhasa terrane provides important constraints on the amount of crustal shortening within Asia and the evolution of the Neo-Tethys Ocean. However, three different models have been proposed for the shape of the Lhasa terrane during the Cretaceous. Analysis of paleomagnetic data from the middle and western segments of the Lhasa terrane has indicated that the terrane was oriented E-W during the Cretaceous (Ma et al., 2014), whereas other studies have proposed that the terrane was oriented NW-SE during this period (Yi et al., 2015; Ma et al., 2022). In contrast, Tong et al. (2019, 2022) proposed that the Lhasa terrane had the shape of an inverted triangle during the Cretaceous.

Most of the existing Cretaceous paleomagnetic results for the Lhasa terrane have been obtained from its middle and western segments. These paleomagnetic data suggest that the middle and western segments of the Lhasa terrane were respectively located at ~10°N and ~18°N paleolatitude during the Cretaceous (e.g., Cao et al., 2017a; Bian et al., 2020; Li et al., 2022; Ma et al., 2022; Tong et al., 2022). Different vertical-axis rotations affected different parts of the Lhasa terrane following the India-Asia collision (Yang et al., 2015b; Cao et al., 2017a; Chen et al., 2017; Ma et al., 2017; Tong et al., 2017; Meng et al., 2018). The available Cretaceous paleopoles of the Lhasa terrane indicate different vertical-axis rotations between different sampling sites within the same segment, and therefore there are differences in the measured paleomagnetic poles obtained by these previous studies. These measured paleomagnetic poles may deviate from the apparent polar wander path of the Lhasa terrane. Hence, paleomagnetic data from the western and middle segments of the Lhasa terrane cannot necessarily be used to directly constrain the paleolatitudes of its eastern segment. Therefore, the most effective method for constraining the Cretaceous paleogeography of the Lhasa terrane is to determine the paleolatitudes of different segments of the terrane. However, insufficient paleomagnetic data are available to accurately constrain the Cretaceous paleolatitude of the eastern segment of the Lhasa terrane. Paleomagnetic data obtained by Zhu et al. (1981) were derived from a small number of samples of Cretaceous sandstone and granodiorite from the eastern segment of the Lhasa terrane, and they lacked accurate geochronological constraints or field tests. Tong et al. (2017) obtained a Late Cretaceous paleolatitude of ~10.5°N from red beds on the northeastern margin of the Lhasa terrane, but the geological age of these red beds is poorly constrained, as they lack a precise isotope chronology or age-diagnostic fossils.

In this study, we carried out new paleomagnetic and geochronological investigations of Early Cretaceous volcanic rocks from the eastern segment of the Lhasa terrane. The results constrain the Early Cretaceous paleogeographic position and shape of the Lhasa terrane, enabling estimates to be made of the amount of crustal shortening in Asia, and they advance our understanding of the evolution of the Neo-Tethys Ocean.

GEOLOGICAL SETTING AND SAMPLING

The Lhasa terrane is located in the southern part of the Tibetan Plateau and extends ~2500 km E-W. The Bangong-Nujiang suture zone separates the Lhasa terrane from the Qiangtang terrane to the north, and the Indus–Yarlung Zangbo suture zone separates the Lhasa terrane from the Tethyan Himalaya terrane to the south (Fig. 1). The study area for this investigation was located in Luolong County, in the eastern segment of the Lhasa terrane. The strata in this study area are mostly Jurassic and Cretaceous in age (Fig. 2A). The Jurassic strata comprise the Middle Jurassic Xihu and Sangkalayong Formations and the Middle to Upper Jurassic Lagongtang Formation. The Xihu Formation consists mainly of carbonaceous slate, siltstone, metamorphic quartz sandstone, and slate, whereas the Sangkalayong Formation consists mainly of bioclastic limestone and gray-black shale. The Cretaceous strata include the Lower Cretaceous Duoni and Bianba Formations and the Upper Cretaceous Zonggei Formation (Bianba region, Geology Survey of Tibet, 2005a). The Duoni Formation consists of siltstone and gray-white quartz sandstone, and the Bianba Formation consists of mudstone, siltstone, and dolomite. The Zonggei Formation is dominated by conglomerate, sandstone, shale, and volcanic rocks.

The sampled section is located ~2 km south of Luolong County (30.7°N, 95.9°E; Fig. 2B). The section comprises intercalated conglomerate, sandstone, limestone, and volcanic rocks, in particular, gray-green and gray-purple andesite and

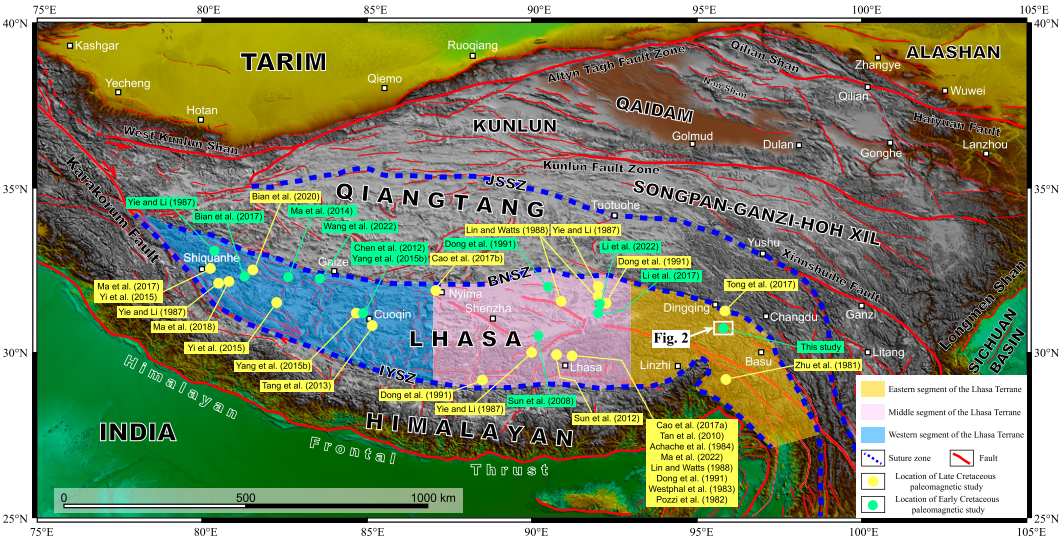


Figure 1. Tectonic map of the Tibetan Plateau, showing the locations of samples collected from the Cretaceous units of the Lhasa terrane in previous paleomagnetic studies. IYSZ—Indus–Yarlung Zangbo suture zone; BNSZ—Bangong-Nujiang suture zone; JSSZ—Jinsha suture zone.

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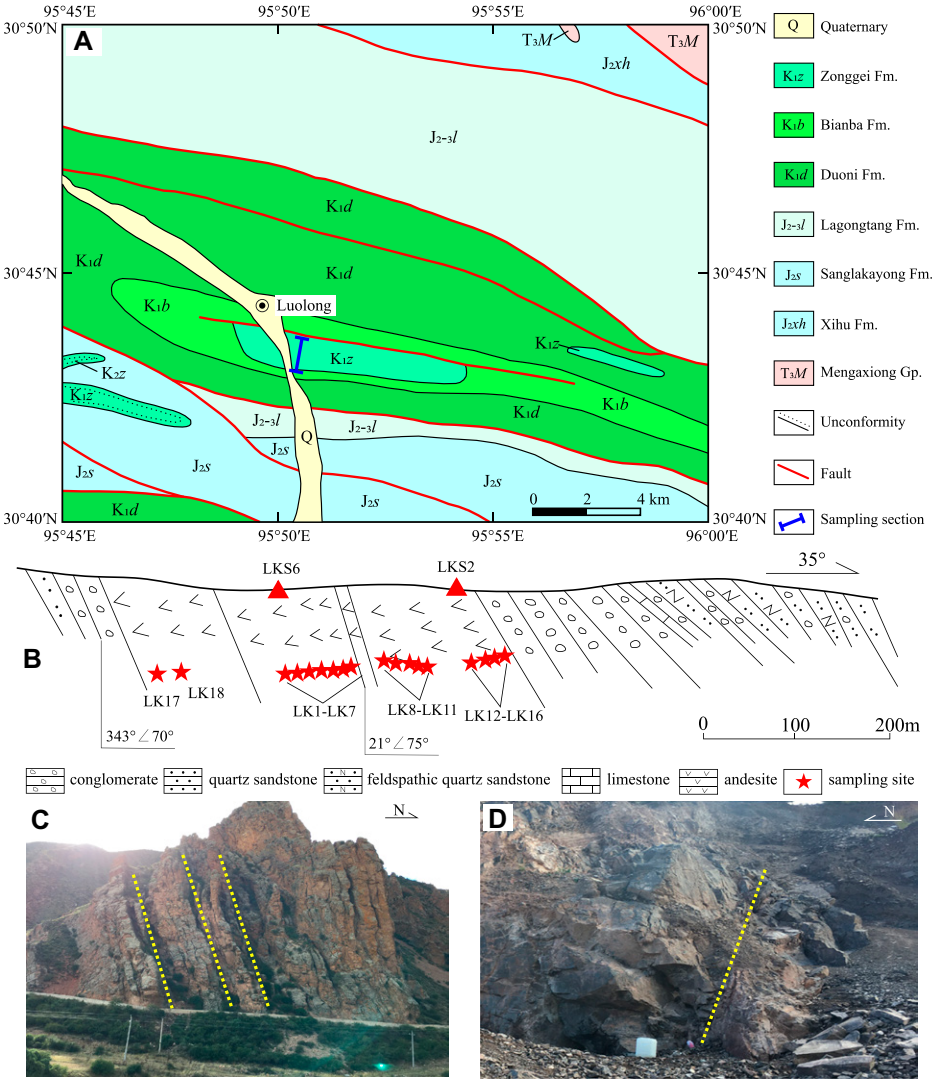


Figure 2. (A) Geological map of the study area and (B) simplified geological cross section of the sampling area, modified from the Bianba regional geological investigation report by the Geology Survey of Tibet (2005a). (C–D) Outcrop photographs, showing (C) the bedding of the conglomerates underlying the andesite, and (D) the lava boundary.

brecciated andesitic lava. The contact between these Early Cretaceous strata and the overlying Paleocene units is unconformable throughout the Lhasa terrane (Naqu region, Geology Survey of Tibet, 2005b). The Early Cretaceous units are folded, and the Paleocene units have been folded into a monocline. These observations raise the possibility that the Early Cretaceous strata of the eastern segment of the Lhasa terrane were folded during the Late Cretaceous.

Drill cores for paleomagnetic analyses were collected from the gray-purple andesite of the Zonggei Formation. These volcanic rocks were previously thought to be of Late Cretaceous age (Bianba region, Geology Survey of Tibet, 2005a), but our new isotope geochronological results indicate that they were formed ca. 127–124 Ma (see the section titled "U-Pb Zircon Dating"). Similarly, Li et al. (2022) obtained an Early Cretaceous (114–110 Ma) age for the Zonggei Formation volcanic rocks in the Naqu area of the Lhasa terrane.

Approximately 160 cores were collected from 18 sites. The cores were oriented using both sun and magnetic compasses. The contact between the lava and the bedding in the underlying conglomerate was used to determine the attitude of the volcanic rocks. Two sets of samples were also collected for geochronological analyses, and two fresh andesite samples (LKS2 and LKS6) were taken from the upper and lower parts of the studied section, respectively.

The methods employed for petrographic observations, U-Pb zircon dating, rock magnetism, and paleomagnetism investigations are described in the Supplemental Material[1].

RESULTS

U-Pb Zircon Dating

Zircon crystals in the studied volcanic rocks were generally euhedral to subhedral with prismatic shapes and 50–100 μm in size (Figs. 3A and 3B). The zircons exhibited oscillatory zoning in cathodoluminescence (CL) images, indicating a magmatic origin. In total, 41 zircon grains were analyzed from samples LKS2 and LKS6 by laser ablation–inductively coupled plasma–mass spectrometry (Table S1). The $^{206}Pb/^{238}U$ concordia ages for samples LKS2 and LKS6 were 124.01 ± 0.34 Ma (mean square of weighted deviates [MSWD] = 0.15) and 126.8 ± 0.38 Ma (MSWD = 2.1), respectively (Figs. 3C and 3D). These results demonstrate that the volcanic rocks in the studied section were formed during the Early Cretaceous (ca. 125 Ma).

Rock Magnetic Results

The medium and soft magnetic components of sample LK11–5 were unblocked at ~580 °C, and the hard component was unblocked at ~680 °C (Fig. 4A). The saturated isothermal remanence magnetization (SIRM) acquisition curve illustrates that the magnetization of sample LK11–5 increased rapidly below 200 mT and approached saturation at ~300 mT (Fig. 4C). Its magnetization decreased to zero at ~60 mT during back-field demagnetization of the SIRM (Fig. 4C). The hysteresis loop of sample LK11–5 was saturated by ~300 mT, which shows that the magnetic behavior of the sample is controlled mainly by low-coercivity magnetic grains (Fig. 4E). The first-order reversal curve (FORC) for this sample displays symmetrical contours with low coercivity and pseudo-single-domain (PSD) domain characteristics (Fig. 4G). These results indicate that magnetite and a small amount of hematite are the primary magnetic carriers in sample LK11–5.

The soft component of sample LK17–8 was unblocked at ~580 °C, whereas the medium and hard components were unblocked at ~680 °C (Fig. 4B). The SIRM acquisition curve of LK17–8 was not saturated at 1 T, and its magnetization decreased to zero at ~240 mT during back-field demagnetization (Fig. 4D). The hysteresis loop of sample LK17–8 shows

[1]Supplemental Material. Methods and Tables S1–S3. Please visit https://doi.org/10.1130/GSAB.S.21971054 to access the supplemental material, and contact editing@geosociety.org with any questions.

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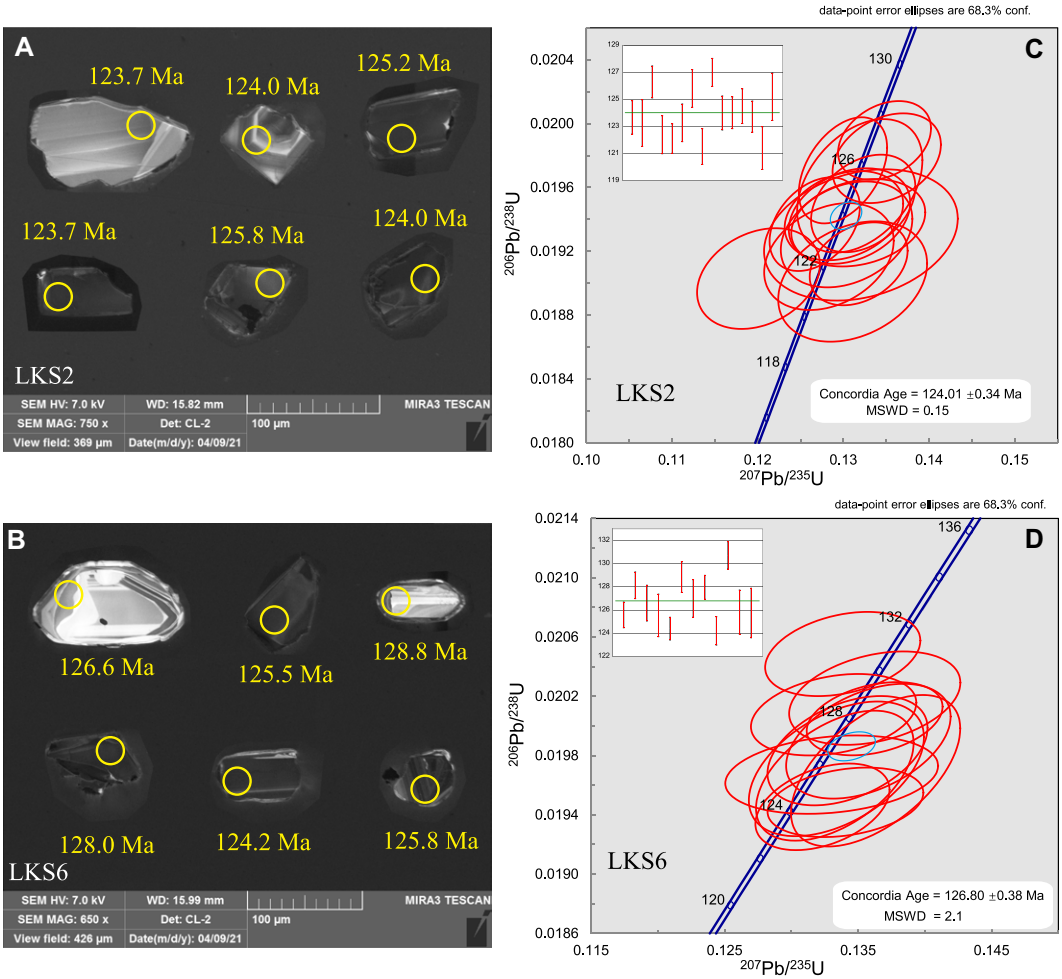


Figure 3. (A–B) Cathodoluminescence images of zircon grains from the Early Cretaceous volcanic rocks, showing the locations of $^{206}Pb/^{238}U$ analyses and age data, and (C–D) U-Pb concordia diagrams. MSWD—mean square of weighted deviates.

a goose-neck shape and was unsaturated at 1 T (Fig. 4F). The FORC diagram for sample LK17–8 has contours compressed along the coercivity (Hc) axis (Fig. 4H), characteristic of single-domain (SD) grains. These results suggest that hematite and magnetite are the primary magnetic carriers in sample LK17–8.

In summary, the rock magnetic results indicate that the primary magnetic carriers of the volcanic rocks in this study are hematite and magnetite.

SEM Observations

Scanning electron microscope (SEM) observations and energy-dispersive X-ray spectroscopic (EDS) analyses were carried out on a representative sample. SEM observations revealed subhedral to euhedral magnetic grains mainly of a few micrometers to ~200 μm in size and showing clear boundaries with the groundmass (Figs. 5A–5C). Their composition, which is dominated by iron and oxygen (Figs. 5D and 5E), indicates that the gains are magnetite or hematite. The magnetic grains do not have oxidized rims (Figs. 5B and 5C). These observations indicate that the magnetic grains have not been altered by chemical fluids or weathering and thus they may carry a primary remanent magnetization.

Paleomagnetic Results

The thermal demagnetization results were subjected to principal component analysis (Kirschvink, 1980), and the mean directions were calculated using Fisher statistics (Fisher, 1953), using the computer programs kirsch.exe (Enkin, 1990) and Paleomac (Cogné, 2003).

In the thermal demagnetization data, it was possible to isolate a low-temperature component (LTC) between room temperature and 300 °C (Figs. 6A–6K and 6Q; Table S2). The mean direction of the LTC isolated from the 89 samples is declination (Dg) = 12.4°, inclination (Ig) = 53.8°, $\kappa g = 54.4$, and $\alpha_{95} = 2.1°$ (κ—best estimate of the precision parameter; α_{95}—radius of the 95% probability ellipse around the mean direction) (in geographic coordinates), and $Ds = 15.9°$, $Is = -20.6°$, $\kappa s = 53.0$, and $\alpha_{95} = 2.1°$ (in stratigraphic coordinates) (Fig. 7A). The LTC direction before tilt correction is statistically similar to the present geomagnetic field direction (PGF), which has a declination of 359.7° and an inclination of 47.6° at the sample site (30.8°N, 95.9°E). This result indicates that the LTC may be a viscous remanence of the PGF.

A high-temperature component (HTC) was isolated between ~420 °C and 580 °C for samples in which magnetite is the primary magnetic carrier (Figs. 6A–6E, 6G–6K, and 6R; Table S3). In samples with hematite as the primary magnetic carrier, the HTC was isolated between ~420 °C and 680 °C (Figs. 6F and 6L–6P). The thermal demagnetization results are consistent with the rock magnetic data, implying that the HTCs are carried by hematite or magnetite. The HTCs decay linearly toward the origin. The thermal demagnetization results also suggest that the hematite and magnetite have similar HTC directions (Fig. 6). The mean HTC direction for the 17 sites is $Dg = 95.4°$, $Ig = 89.4°$, $\kappa g = 19.5$, and $\alpha_{95} = 8.3°$ in geographic coordinates and

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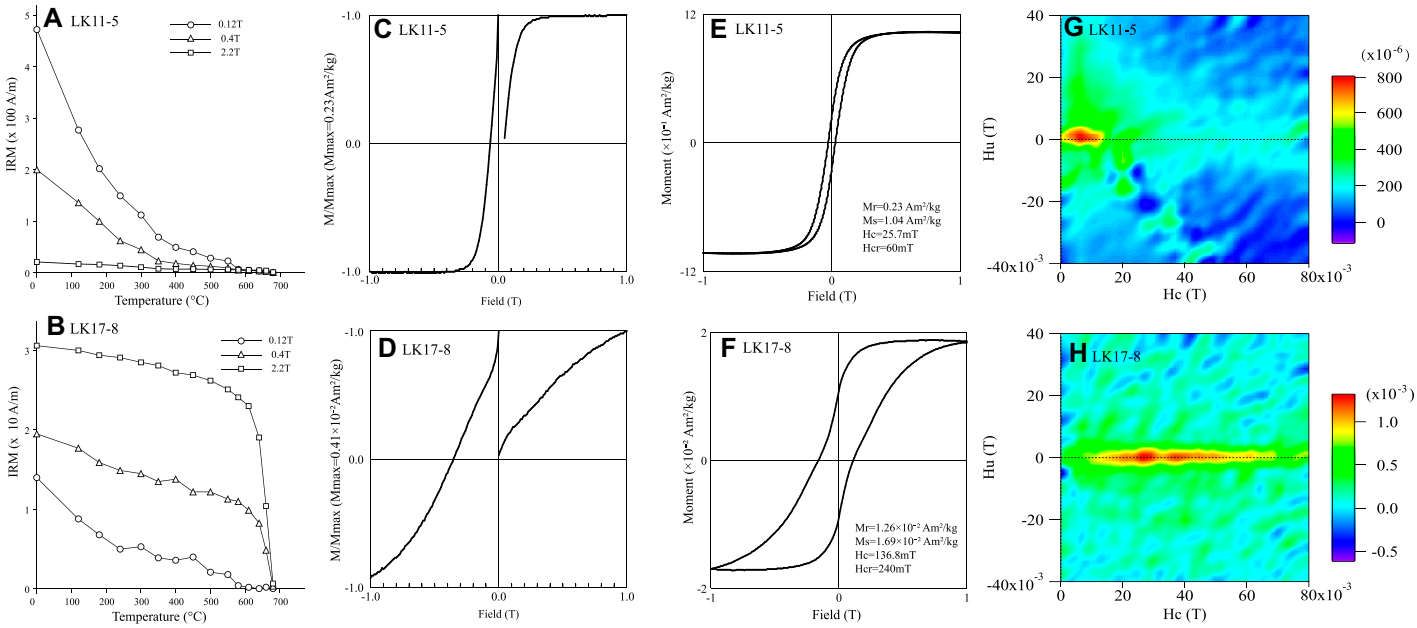


Figure 4. Rock magnetic results for representative samples: (A–B) thermal demagnetization of a three-component isothermal remanence magnetization (IRM), (C–D) saturated isothermal remanence magnetization (SIRM) acquisition and back-field demagnetization curves, (E–F) hysteresis loops, and (G–H) first-order reversal curve (FORC) diagrams. M—magnetization; Mmax—maximum magnetization; Mr—saturation remanent magnetization; Ms—saturation magnetization; Hc—coercive force; Hcr—remanent coercivity; Hu—magnetostatic interactions.

Ds = 21.8°, *Is* = 19.0°, κs = 65.8, α_{95} = 4.4° in stratigraphic coordinates (Fig. 7B; Table 1). The mean paleomagnetic pole is 60.9°N, 227.2°E, with *dp/dm* = 2.4°/4.6° (dp/dm—semi-axes of the 95% probability ellipse around the mean pole).

The HTC directions show a greater difference compared to the PGF before tilt correction (Fig. 7B) and are more clustered (*ks/kg* = 3.37) in stratigraphic coordinates than in geographic coordinates. Thus, the paleomagnetic data pass the fold test of McElhinny (1964). The

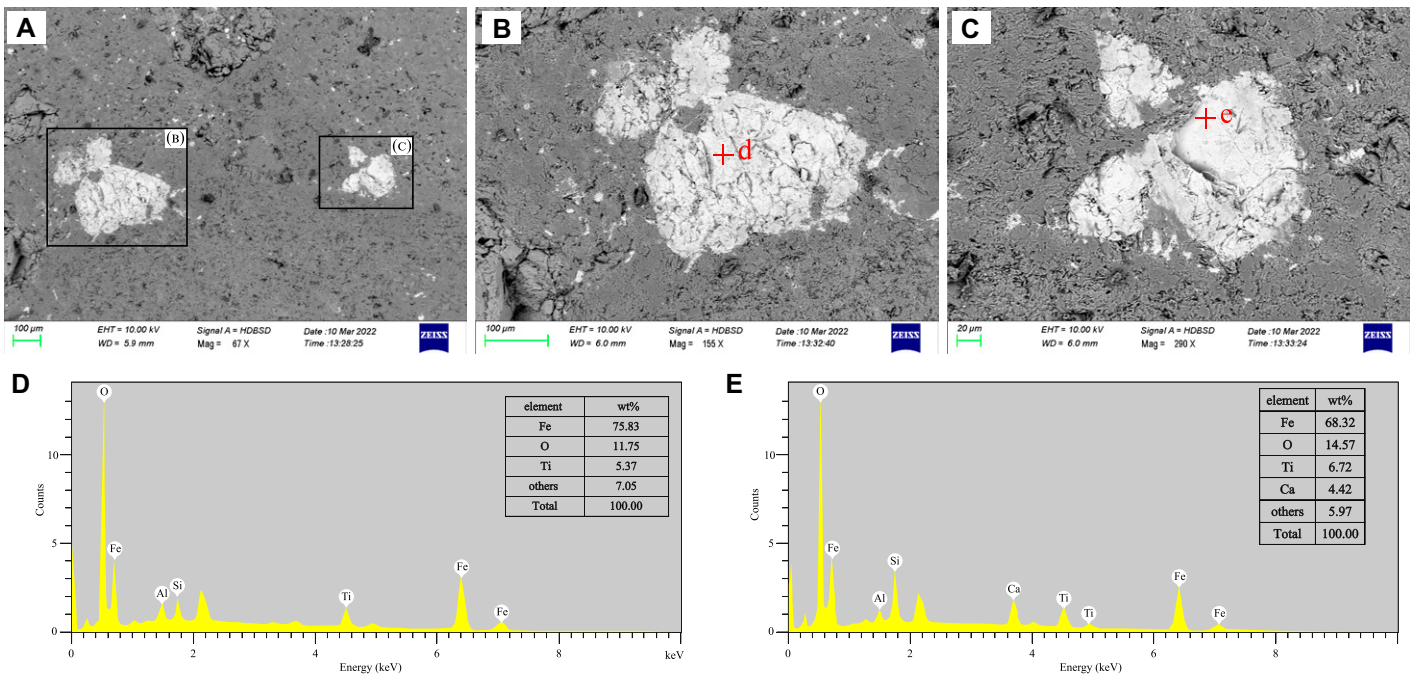


Figure 5. (A–C) Scanning electron microscope (SEM) backscattered electron images of magnetite grains and (D–E) energy-dispersive spectroscopy (EDS) spectra derived from the sites shown in A–C.

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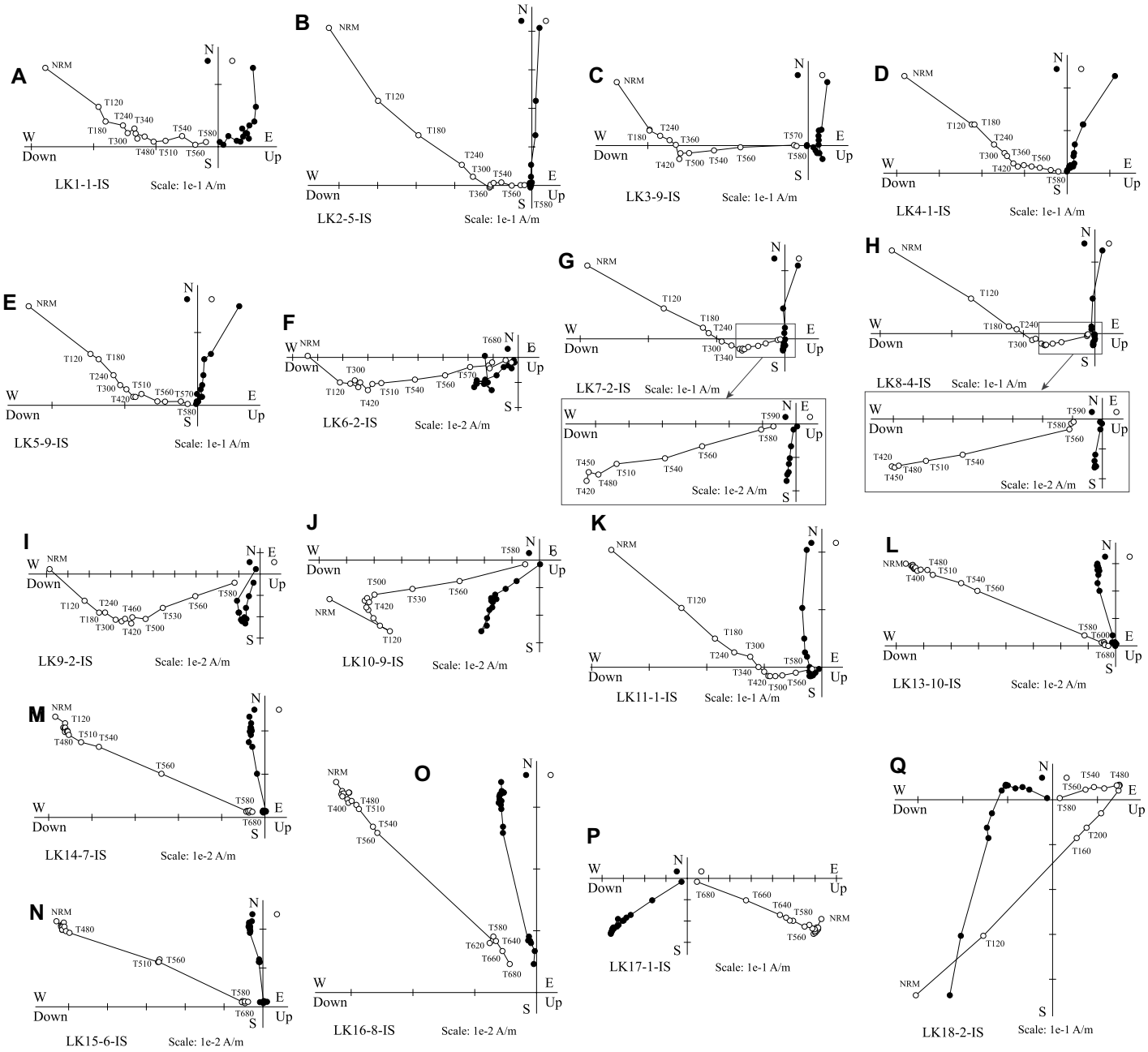


Figure 6. Demagnetization diagrams for representative samples. Diagrams are plotted in geographic coordinates. Closed symbols represent horizontal projections, and open symbols represent vertical projections. Demagnetization steps are in °C in all cases. NRM—natural remanent magnetization; T—temperature.

paleomagnetic data also pass both the fold test (McFadden, 1990) and the reversal test (McFadden and McElhinny, 1990), supporting the conclusion that they represent a primary magnetization. The typical magmatic structures of the volcanic rocks support the hypothesis that the magnetite and hematite grains have not been altered after diagenesis, and thus they carry the primary remanent magnetization.

The analyzed samples were collected from multistage lava flows. Isotope geochronological results indicate that these lavas formed during the period ca. 127–124 Ma. The HTC directions include antipodal dual polarities. Samples from sites LK1–LK16 showed normal polarity, and samples from sites LK17 and LK18 showed reversed polarity. The U-Pb dating results indicate that the studied volcanic rocks were formed over an interval of >3 m.y. In addition, the mean HTC direction with $A_{95} = 1.7°$ for 142 samples falls within the reliability envelope of 1.6°–3.6° (Deenen et al., 2011). These results indicate that paleosecular variation (PSV) is averaged out in the mean HTC direction that was determined in this study.

The combined results of the fold test, reversal test, PSV test, and microstructural observations together indicate that the HTC direction obtained in this study represents the primary remanent magnetization of the Early Cretaceous (ca. 125 Ma) volcanic rocks of the eastern segment of the Lhasa terrane.

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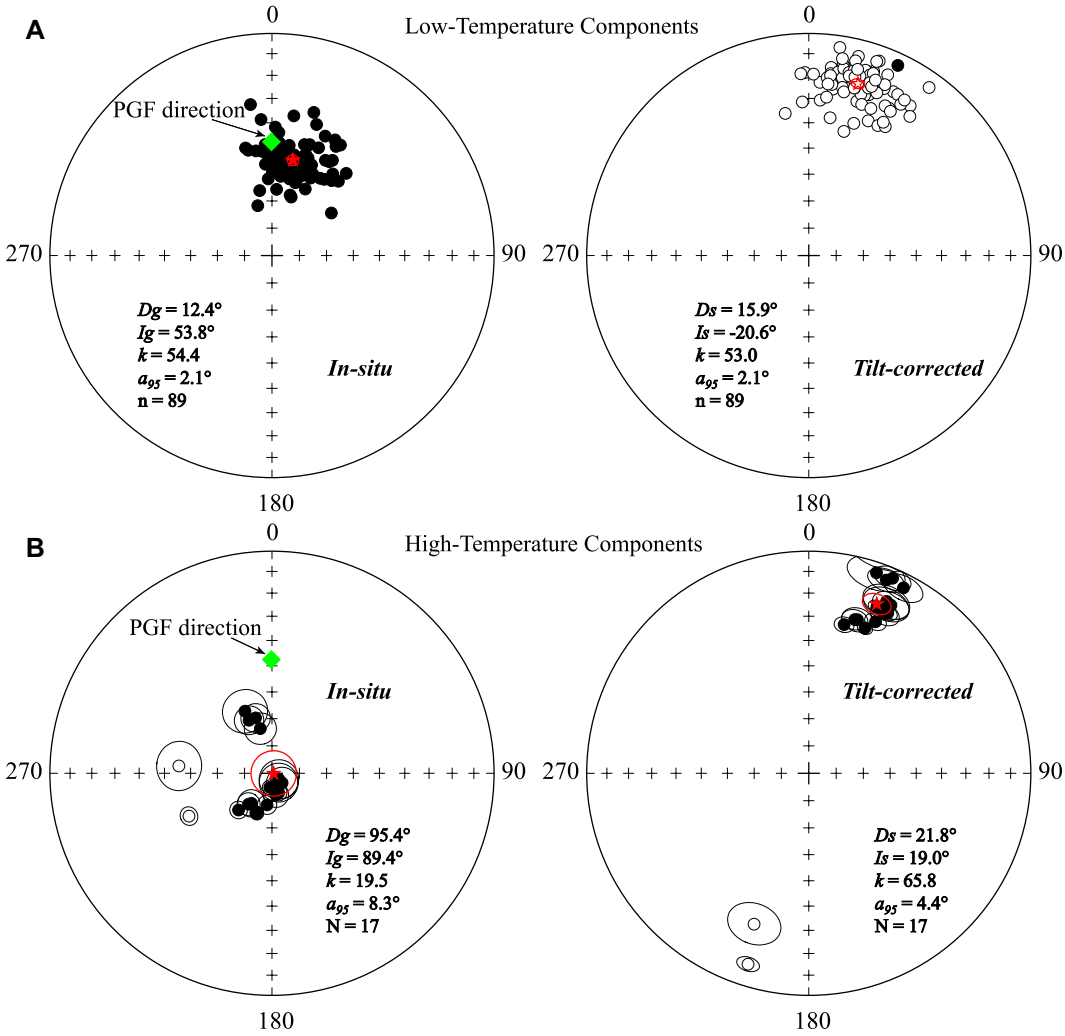


Figure 7. Equal-area projections of the mean directions. Red stars indicate the mean direction for (A) 89 samples and (B) 17 sites. Open symbols denote upward inclinations, and solid symbols denote downward inclinations. PGF—present geomagnetic field.

TABLE 1. SITE-MEAN DIRECTIONS FROM THE EARLY CRETACEOUS VOLCANIC ROCKS OF THE EASTERN SEGMENT OF THE LHASA TERRANE

Site	Strike/dip (°)	*n*/*N*	*Dg* (°)	*Ig* (°)	*Ds* (°)	*Is* (°)	κ	α_{95} (°)	PLat. (°N)	PLong. (°E)	*dp*/*dm* (°)
LK1	291/75	10/10	133.5	84.9	26.0	16.9	73.3	5.7	57.2	222.7	3/5.9
LK2	291/75	10/10	188.5	84.8	22.2	20.1	321.0	2.7	61.0	225.8	1.5/2.8
LK3	291/75	8/8	160.8	82.5	26.2	20.7	349.2	3	58.5	219.7	1.7/3.2
LK4	291/75	8/8	131.6	86.6	24.3	16.2	60.9	7.2	58.1	225.5	3.8/7.4
LK5	291/75	6/6	144.4	85.8	24.6	17.3	89.7	7.1	58.3	224.3	3.8/7.3
LK6	291/75	8/8	214.2	76.3	17.5	28.4	122.0	5.0	67.6	226.2	3.0/5.5
LK7	291/75	8/8	188.8	78.3	23.7	26.4	254.0	3.5	62.5	218.2	2.1/3.8
LK8	291/75	8/8	169.6	82.0	25.5	21.7	144.9	4.6	59.4	219.8	2.6/4.9
LK9	291/75	9/9	200.8	74.4	21.1	30.6	429.3	2.5	66.1	217.6	1.6/2.8
LK10	291/75	10/10	222.3	71.8	13.3	31.8	234.3	3.2	71.9	230.9	2.0/3.6
LK11	291/75	11/11	217.1	75.8	16.6	28.6	118.1	4.2	68.3	227.6	2.5/4.6
LK12	311/73	0/11	Not possible to isolate an HTC direction								
LK13	311/73	9/9	336.6	68.8	21.8	7.0	94.1	5.3	55.9	234.5	2.7/5.3
LK14	311/73	8/8	345.4	73.1	27.0	7.0	91.3	5.8	52.7	227.4	2.9/5.8
LK15	311/73	9/9	343.5	68.8	23.2	5.0	94.9	5.3	54.3	233.5	2.7/5.3
LK16	311/73	4/10	336.5	65.1	18.6	5.3	129.6	8.1	56.9	240.2	4.1/8.1
LK17	253/70	11/11	242.7	−55.4	197.7	−10.7	198.2	3.3	−59.7	59.0	1.7/3.3
LK18	253/70	5/5	274.5	−55.4	200.0	−28.5	80.0	8.6	−66.0	41.7	5.2/9.4
Mean		17(142)	95.4	89.4	21.8	19.0	19.5 65.8	8.3 4.4			

Notes: *n*/*N*—number of samples used to calculate mean directions/total number of samples demagnetized; *Dg*, *Ig*—declination and inclination in geographic coordinates; *Ds*, *Is*—declination and inclination in stratigraphic coordinates; κ—best estimate of the precision parameter; α_{95}—radius of the 95% probability ellipse around the mean direction; PLat. and PLong.—latitude and longitude of paleomagnetic poles; *dp*/*dm*—semi-axes of the 95% probability ellipse around the mean pole; A_{95}—radius of the 95% probability ellipse around the mean pole; HTC—high-temperature component. Fold test: Xi1 = 7.38 in situ, Xi1 = 8.99 after tilt correction, Xi2 = 14.27 in situ, Xi2 = 3.89 after tilt correction, Xi = 4.8 at the 95% confidence level, Xi = 6.72 at 99% confidence level (McFadden, 1990). Reversal test: Statistics (*r*) = 3.3° < critical gamma (*rc*) = 14.2°, passes the reversal test with a C classification at the 95% confidence level (McFadden and McElhinny, 1990).

DISCUSSION

Early Cretaceous Paleolatitude of the Lhasa Terrane

Despite the numerous paleomagnetic studies conducted in the Lhasa terrane, the Early Cretaceous paleolatitude of the eastern segment of the Lhasa terrane remains poorly constrained. We adopted the criteria of Meert et al. (2020) to evaluate the reliability of the Cretaceous paleomagnetic data obtained in previous studies. Paleomagnetic data with an *R* value <5 were excluded, and they are not discussed here. In addition, because different vertical-axis rotations affected different parts of the Lhasa terrane following the India-Asia collision (Yang et al., 2015b; Cao et al., 2017a; Chen et al., 2017; Ma et al., 2017; Tong et al., 2017; Meng et al., 2018), paleomagnetic data from individual sites within the terrane cannot necessarily be used directly to constrain the paleolatitudes of other areas. To identify spatial differences, the paleomag-

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TABLE 2. EARLY CRETACEOUS PALEOMAGNETIC DATA FROM THE LHASA TERRANE AND TETHYAN HIMALAYA TERRANE

Latitude (°N)	Longitude (°E)	Age (Ma)	Lithology	*N/n*	Plat. (°N)	Plong. (°E)	A_{95} (*dp/dm*) (°)	Paleolatitude (°N)	*R* value	Reference
Eastern segment of the Lhasa terrane (reference point: 30°N, 96°E)										
30.7	95.9	~125	Volcanic rocks	17/142	60.9	227.2	2.4/4.6	9.2 ± 2.4	123F5R7(7)	This study
Middle segment of the Lhasa terrane (reference point: 30°N, 91°E)										
31.5	92.1	~112	Volcanic rocks	9/59	72.0	252.6	6.7	12.8 ± 6.7	123□5□7(5)	Li et al. (2022)
31.3	91.9	~120	Volcanic rocks	19/139	66.9	281.2	6.1	7.2 ± 6.1	123F5□7(6)	Li et al. (2017)
30.5	90.1	~114	Volcanic rocks	15/88	66.4	220.3	6.9/9.3	13.8 ± 6.9	123□5□7(5)	Sun et al. (2008)
Paleolatitude fitted by the small circle method								11.3 ± 8.8		
32.0	*90.5*	*K_1*	*Sandstone*	*2/6*	*–10.5*	*17.9*	*7.3/14.0*	*9.0 ± 7.3*	*1□□□□□7(2)*	*Dong et al. (1991)*
30.0	*91.0*	*K_1*	*Red beds*	*14/110*	*63.4*	*323.2*	*4.8*	*12.1 ± 4.8*	*□2□F5□7(4)*	*Achache et al. (1984)*
Western segment of the Lhasa terrane (reference point: 31°N, 82°E)										
32.3	82.6	~132–120	Volcanic rocks	51/444	61.4	192.9	2.1	17.8 ± 2.1	123F5□7(6)	Ma et al. (2014)
31.3	85.1	~130–110	Volcanic rocks	18/162	58.2	341.9	4.6	21.0 ± 4.6	123F5R7(7)	Chen et al. (2012)
31.2	84.4	~121–117	Volcanic rocks	12/116	70.5	292.9	7.4	13.9 ± 7.4	123□5□7(5)	Yang et al. (2015b)
32.2	80.4	~116–113	Volcanic rocks and limestone	19/205	69.1	319.8	4.8	18.6 ± 4.8	123F5□7(6)	Bian et al. (2017)
32.4	83.4	~120–106	Red beds	56/453	81.6	265.9	1.2/2.0	22.6 ± 1.2	123F5□7 (6)	Wang et al. (2022)
Paleolatitude fitted by the small circle method								18.8 ± 4.2		
33.1	*80.3*	*K_1*	*Sandstone*	ND/ND	*48.3*	*335.4*	*4.4/8.0*	*12.8 ± 4.4*	*1□□□□□7(2)*	*Yie and Li (1987)*
Middle-eastern segment of the Tethyan Himalaya terrane (reference point: 30°N, 91°E)										
28.5	*87.0*	*~120–149*	*Sandstone*	ND/*201*	*4.4*	*255.9*	*4.6/5.2*	*–52.7 ± 4.6*	*1□3□5□7(4)*	*Huang et al. (2015)*
28.6	*89.3*	*K_1*	*Limestone*	*3/11*	*30.0*	*285.3*	*6.5/9.9*	*–28.5 ± 6.5*	*□□□□5□7(2)*	*Dong et al. (1991)*
28.8	*86.3*	*K_1*	ND	ND/*56*	*–32.7*	*115.7*	*5.1/8.4*	*–23.1 ± 5.1*	*□□□□5□7(2)*	*Li et al. (2006)*
28.8	91.3	~135–124	Volcanic rocks	26/216	–5.9	308.0	6.1	–47.7 ± 6.1	123F5R7(7)	Ma et al. (2016)
28.9	91.3	~138–135	Volcanic rocks	31/219	0.9	293.4	7.0	–52.4 ± 7.0	123F5R7(7)	Bian et al. (2019)
28.1	92.4	~134–131	Volcanic rocks	31/225	–26.8	315.2	5.7	–51.2 ± 5.7	123F5R7(7)	Yang et al. (2015a)
28.8	91.1	~134	Basalt	11/185	6.3	308.6	9.1	–38.8 ± 9.1	123F5□7(6)	Zhang et al. (2019)
28.1	92.4	137–125	Volcanic rock and limestone	28/194	22.0	266.7	7.6	–37.8 ± 7.6	123F5R7(7)	Meng et al. (2019)
Western segment of the Tethyan Himalaya terrane (reference point: 31°N, 82°E)										
28.8	*83.8*	*~121–112*	*Sediment*	ND/*95*	*12.0*	*289.0*	*6.0/7.5*	*–39.8 ± 6.0*	*1□3□5□7(4)*	*Klootwijk and Bingham (1980)*
29.7	*84*	*~120–94*	*Chert*	ND/*53*	*25.0*	*285.7*	*4.8/7.1*	*–29.6 ± 4.8*	*1□3□5R7(4)*	*Qin et al. (2019)*
29.7	83.6	125–134	Sandstone and chert	ND/53	29.7	260.1	4.9	–29.3 ± 4.9	1□3F5R7(6)	Meng et al. (2019)

Notes: *N/n*—number of sites/number of samples used to calculate Fisher mean; Plat./Plong.—latitude/longitude of paleomagnetic poles; A_{95}—radius of the 95% confidence level of the paleomagnetic pole; K_1—Early Cretaceous; ND—no data. Paleolatitude: positive numbers are Northern Hemisphere paleolatitudinal positions; negative numbers are Southern Hemisphere paleolatitudinal positions. *R* value—data quality criteria (number of criteria met) after Meert et al. (2020); '□' in the criterion column fails to fulfill this criterion. Paleomagnetic data with *R* values <5, which are shown in italics, were not employed to calculate paleolatitudes.

netic data were therefore separated into three groups based on geographical location, namely, the western (longitude range ~80°E to 87°E), middle (longitude range 87°E to 93°E), and eastern (longitude range 93°E to ~99°E) segments of the Lhasa terrane. We calculated the paleolatitude for a reference point within each of the three segments, at 31°N, 82°E and 30°N, 91°E and 30°N, 96°E. The reference points are along the Indus–Yarlung Zangbo suture zone, which we take to represent the southern margin of the Lhasa terrane.

For the eastern segment of the Lhasa terrane, we prioritized our new paleomagnetic results (R value = 7) from the ca. 125 Ma volcanic rocks, for which the paleomagnetic direction is $Ds = 21.8°$, $Is = 19.0°$, $\kappa s = 65.8$, and $\alpha_{95} = 4.4°$, $N = 17$ in stratigraphic coordinates. The mean paleomagnetic pole is 60.9°N, 227.2°E with $dp/dm = 2.4°/4.6°$. These new results define the Early Cretaceous paleolatitude of the southern margin of the eastern Lhasa terrane as 9.2 ± 2.4°N.

For the middle segment of the Lhasa terrane, a small circle fit to the three most reliable Early Cretaceous paleomagnetic data reported by Sun et al. (2008) and Li et al. (2017, 2022) indicates a colatitude of 78.7° ± 8.8°, which corresponds to a paleolatitude of 11.3 ± 8.8°N. For the western segment of the Lhasa terrane, five previously reported paleomagnetic data have R values ≥5 (Table 2). The paleomagnetic data obtained by Wang et al. (2022) may have been influenced by syntectonic sedimentation; therefore, we used the paleomagnetic pole after syntectonic sedimentation correction of the paleomagnetic data reported by Wang et al. (2022) for further discussion. A small circle fit to the five reliable Early Cretaceous paleomagnetic data reported by Chen et al. (2012), Ma et al. (2014), Yang et al. (2015b), Bian et al. (2017), and Wang et al. (2022) from the western segment of the Lhasa terrane provides a colatitude for the reference point of 71.2° ± 4.2°, which corresponds to a paleolatitude of 18.8 ± 4.2°N.

This analysis gives Early Cretaceous paleolatitudes of 9.2 ± 2.4°N, 11.3 ± 8.8°N, and 18.8 ± 4.2°N for the eastern, middle, and western segments of the Lhasa terrane, respectively (Fig. 8). The difference in paleolatitude between the eastern and western segments is at least 3.0°, considering the error ranges. These results indicate that the Lhasa terrane as a whole was oriented WNW-ESE during the Early Cretaceous. In addition, the paleolatitudes of the eastern and middle segments of the Lhasa terrane are not significantly different (Fig. 8B). This finding may indicate that the eastern-middle segments of the Lhasa terrane were oriented nearly E-W during the Early Cretaceous.

Seismic tomography of the deeper mantle can reveal the remnants of lithospheric material of a subducted slab and can be used to restore the paleogeography of the plate (Van der Voo et al., 1999). This method has been used to constrain the structure of the southern margin of Asia for the period before the India-Asia collision (Yi et al., 2015; Ma et al., 2022). Evidence from seismic tomography indicates that the remnant lithospheric slab of the Neo-Tethys Ocean has a linear orientation (Van der Voo et al., 1999). The southern margin of the high P-wave velocity anomaly is presently located at ~15°N at 82°E, ~9°N at 91°E, and ~6°N at 96°E (Van der Voo et al., 1999). These results indicate that

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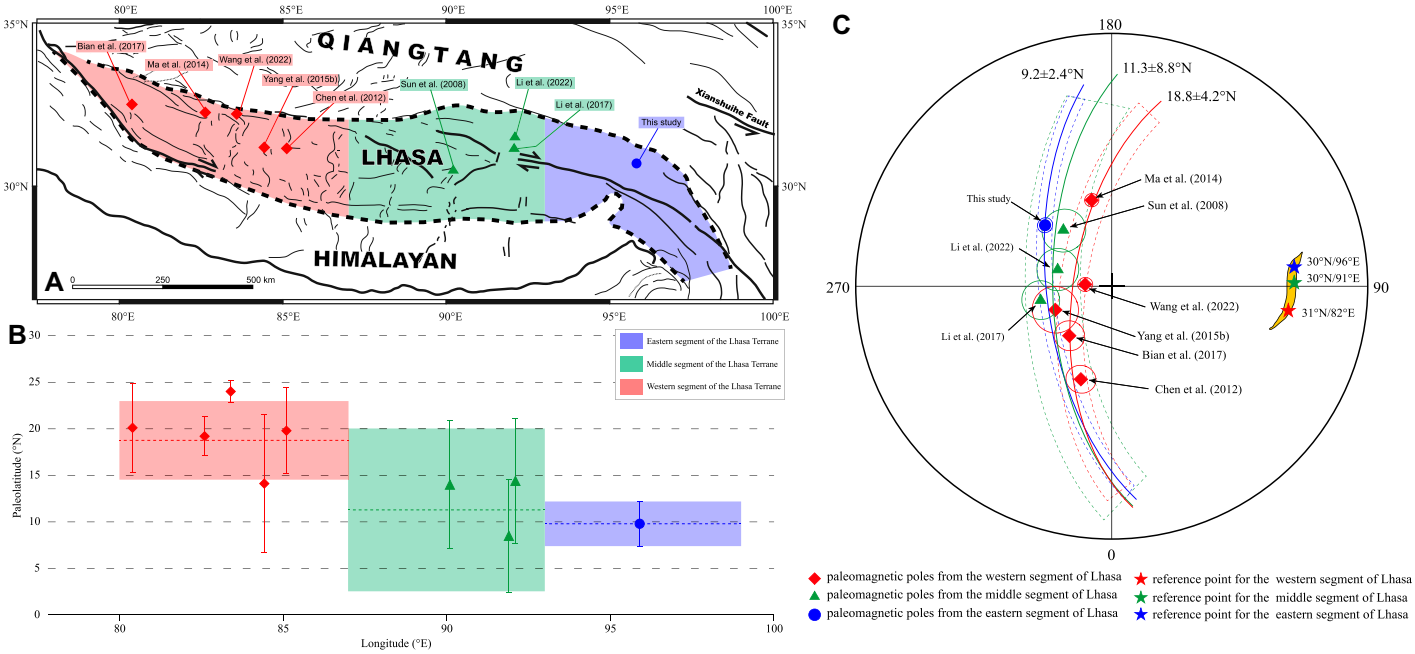


Figure 8. (A) Sampling locations for Early Cretaceous paleomagnetic studies of the Lhasa terrane. (B) Observed and averaged Early Cretaceous paleolatitudes of the different segments of the Lhasa terrane. (C) Early Cretaceous paleomagnetic poles of the Lhasa terrane. Numeric values and references are presented in Table 2. Thicker light-blue lines show the measured paleolatitudes (with 95% confidence level) calculated for the reference point 30°N, 96°E in the eastern segment of the Lhasa terrane. Thicker light-green lines show the measured paleolatitudes (with 95% confidence level) calculated for the reference point 30°N, 92°E in the middle segment of the Lhasa terrane. Thicker light-red lines show the measured paleolatitudes (with 95% confidence level) calculated for the reference point 31°N, 82°E in the western segment of the Lhasa terrane.

the southern margin of Asia had a linear structure prior to the India-Asia collision. Geological evidence indicates that large amounts of N-S crustal shortening (along 94°E) have occurred within Asia since the collision between India and Asia (Li et al., 2015). The crustal shortening has been accommodated mainly by large-scale fold-and-thrust belts within Lhasa, Qiangtang, Songpan-Ganzi, and Qaidam-Qilian-Nanshan (Li et al., 2015), and by the extrusion of the eastern Tibetan Plateau (Royden et al., 2008). However, there are few large-scale fold-and-thrust belts in the region north of the western segment of the Lhasa terrane (along ~82°E), and the western Tibetan Plateau has not undergone obvious lateral extrusion during the Cenozoic (Han et al., 2019). Hence, there has been much greater crustal shortening to the north of the eastern segment of the Lhasa terrane than to the north of the western segment of the Lhasa terrane since the India-Asia collision. Considering the present-day E-W trend of the Lhasa terrane, the Early Cretaceous paleolatitudes of the western and eastern segments of the Lhasa terrane would be expected to be dissimilar.

In addition, the WNW-ESE orientation of the Lhasa terrane during the Cretaceous is more compatible with the India-Asia collision model provided by the geological evidence. Sedimentary studies of the Paleogene strata on the northern margin of the Tethyan Himalaya indicate that the Indian plate quasi-synchronously collided with Asia along the western and central Tethyan Himalaya (Hu et al., 2015, 2016, 2017). The Tethyan Himalaya was oriented NW-SE prior to the India-Asia collision (Tong et al., 2022). The orientation of the middle-western segments of the Lhasa terrane is approximately consistent with the northern margin of the Tethyan Himalaya. This implies that the Tethyan Himalaya approximately collided parallel with the Lhasa terrane. This collision model is consistent with the sedimentological evidence (Hu et al., 2015, 2016, 2017). The Indian plate underwent counterclockwise rotation after the initial collision between India and Asia, which implies nonuniform crustal shortening within the Lhasa terrane. The amount of N-S crustal shortening of the eastern segment of the Lhasa terrane should be higher than that of the western segment. This process may have resulted in the southeastward extrusion of the crustal material of the Lhasa terrane (Tong et al., 2019) and a change in the geometry of the Lhasa terrane.

Insights into Crustal Shortening within Asia

The apparent polar wander path (APWP) of East Asia reported by Cogné et al. (2013) was used to constrain the expected paleolatitude of the Lhasa terrane. The Early Cretaceous (130–110 Ma) mean paleopole for East Asia is 80.9°N, 197.4°E with $A_{95} = 1.5°$ (Table 3). The expected paleolatitude of 27.8 ± 1.5°N for the reference point 30°N, 96°E in the Lhasa terrane was calculated using the Early Cretaceous mean paleopole for East Asia. The Early Cretaceous paleomagnetic pole for the eastern Lhasa terrane of 60.9°N, 227.2°E ($dp/dm = 2.4°/4.6°$) indicates a paleolatitude of 9.2 ± 2.4°N for the reference point 30°N, 96°E. Thus, there is a discrepancy between the measured and expected Early Cretaceous paleolatitudes of the eastern segment of the Lhasa terrane of 18.5 ± 2.1°. This discrepancy implies that 2050 ± 230 km of N-S crustal

TABLE 3. EARLY CRETACEOUS APWP OF EAST ASIA*

Age (Ma)	Paleopole			
	Latitude (°N)	Longitude (°E)	*k*	A_{95} (°)
100	82.2	205.4	118.1	3.5
110	80.8	199.5	78.9	3.5
120	80.6	192.9	60.4	2.9
130	79.9	193.5	70.0	2.6
Mean	80.9	197.4	3775.0	1.5

Notes: APWP—apparent polar wander path; *k*—best estimate of the precision parameter; A_{95}—radius of the 95% confidence level of the paleomagnetic pole.
*After Cogné et al. (2013).

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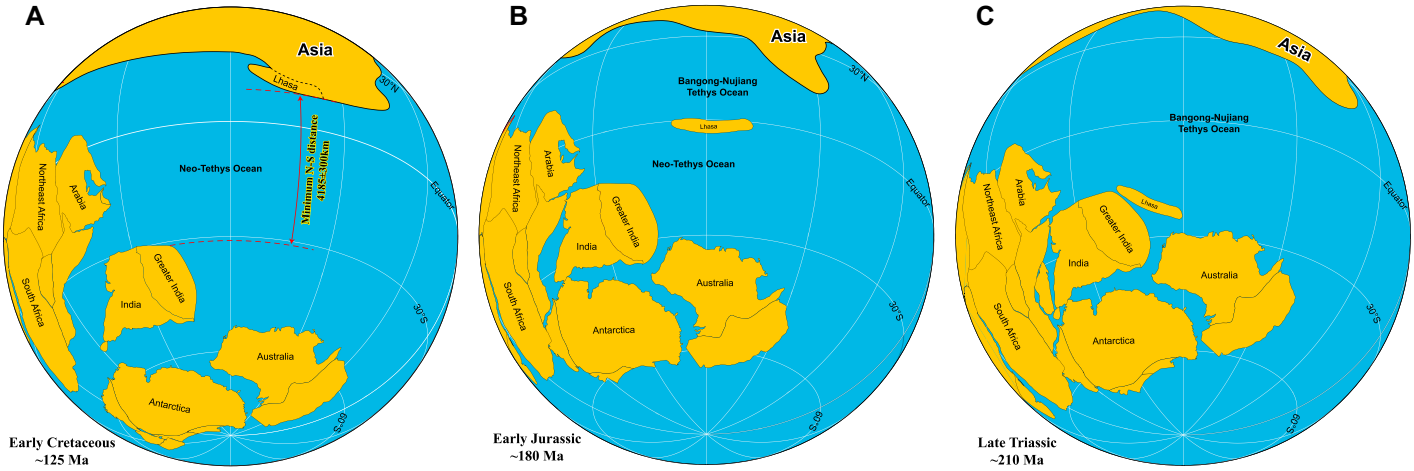


Figure 9. Paleogeographic reconstructions for terranes around the Neo-Tethys Ocean at (A) ca. 125 Ma, (B) ca. 180 Ma, and (C) during the Late Triassic. The data sets for Eurasia, India, Australia, Antarctica, and Africa are from Torsvik et al. (2012), and the paleomagnetic data for the Lhasa terrane ca. 180 Ma and the Late Triassic are from Li et al. (2016) and Zhou et al. (2016), respectively. The shape of Greater India is modified from Ali and Aitchison (2005), Meng et al. (2019), and Zhang et al. (2019). Diagrams were produced using GMAP2015 developed by Trond Torsvik.

shortening has occurred between the Lhasa terrane and stable Asia along 96°E longitude since the Early Cretaceous.

Geological evidence suggests that the Qiangtang terrane has undergone >300 km of crustal shortening since the collision between India and Asia (Kapp et al., 2005, 2007; DeCelles et al., 2007). Mass-balanced cross sections show that the Fenghuoshan and Yushu-Nangqian areas have been shortened by ~43% during the Cenozoic, which indicates ~200 km of crustal shortening in the Songpan-Ganzi terrane (Wang et al., 2002; Spurlin et al., 2005). Based on a comprehensive analysis of the geological evidence, Li et al. (2015) estimated that ~520 km of crustal shortening has occurred in the Qaidam-Qilian-Nanshan region since the Eocene, within which the Qaidam and Qilian-Nanshan areas have undergone ~170 and ~350 km of shortening, respectively. The eastern Tianshan, located near the northwestern Tarim block and the northern Qilian Shan, is thought to record 20–200 km of shortening during the Cenozoic (Yin et al., 1998). In addition, the terranes within the eastern Tibetan Plateau may have been extruded southeastward by ~600 km along strike-slip faults as a result of the India-Asia collision (e.g., Sato et al., 2001; Tapponnier et al., 2001; Otofuji et al., 2010). This extrusion is balanced by ~250 km of N-S crustal shortening (van Hinsbergen et al., 2011). Therefore, the total crustal shortening within Asia since the India-Asia collision is estimated to be ~1300 km. Geological evidence indicates that at least 650 km of additional N-S crustal shortening of the Tibetan Plateau occurred prior to the India-Asia collision as a result of the collision between the Lhasa and Qiangtang terranes (Murphy et al., 1997; Kapp et al., 2003, 2005). During this event, the Lhasa terrane was shortened by ~250 km (Murphy et al., 1997; Volkmer et al., 2007), and the Qiangtang terrane was shortened by ~400 km (Kapp et al., 2005; DeCelles et al., 2007). Therefore, the ~1950 km of crustal shortening estimated from geological evidence is consistent with the present estimate of 2050 ± 230 km based on paleomagnetic constraints.

Tectonic Evolution of the Neo-Tethys Ocean

According to the reconstruction of Gondwana and the Cretaceous drift history of India, the northern margin of the present-day Indian plate was nearly perpendicular to the paleo-equator during the Early Cretaceous (Fig. 9A; Torsvik et al., 2012; Zhang et al., 2019). This configuration means that the Indian plate must have drifted away from Gondwana and undergone large-scale counterclockwise rotation before the India-Asia collision to trend nearly N-S, similar to its present-day orientation. The Tethyan Himalaya terrane was part of Greater India, which comprised part of the Indian plate that was subducted beneath Asia (e.g., Ali and Aitchison, 2005; van Hinsbergen et al., 2012; Meng et al., 2019; Zhang et al., 2019). Reliable paleomagnetic data suggest an Early Cretaceous paleolatitude of ~40°S–50°S for the eastern-middle segments of the Tethyan Himalaya terrane (Table 2; Yang et al., 2015a; Ma et al., 2016; Bian et al., 2019; Meng et al., 2019; Zhang et al., 2019), whereas the Early Cretaceous paleolatitude of the western segment of the Tethyan Himalaya terrane was ~30°S (Meng et al., 2019). This suggests that the northern margin of the present-day Tethyan Himalaya terrane was nearly perpendicular to the paleo-equator during the Early Cretaceous, and therefore the present-day western end point of the Tethyan Himalaya terrane was its northern margin during the Early Cretaceous (Fig. 9A). Consequently, paleomagnetic results from the western segment of the Tethyan Himalaya terrane should be used to constrain the minimum dimension of the Neo-Tethys Ocean.

The paleomagnetic study of Early Cretaceous sandstone and chert in the Zhongba area by Meng et al. (2019) yielded a reliable paleopole for the western segment of the Tethyan Himalaya terrane of 29.7°N, 260.1°E with $A_{95} = 2.8°$. A comparison of the paleomagnetic results of this study with that paleopole suggests a paleolatitudinal difference of 37.7° ± 2.7° between the eastern segment of the Lhasa terrane and the western segment of the Tethyan Himalaya terrane. This result implies that the minimum N-S width of the Neo-Tethys Ocean during the Early Cretaceous (ca. 125 Ma) was 4185 ± 300 km (Fig. 9A).

The separation of the Lhasa terrane from Gondwana during the Late Triassic and its northward drift to equatorial regions by ca. 180 Ma (Li et al., 2016) resulted in the expansion of the Neo-Tethys Ocean. The sedimentary history of the Tethyan Himalaya terrane supports the separation of the Lhasa terrane from Gondwana during the Late Triassic (Liu and Einsele, 1994). The occurrence of Jurassic–Cretaceous magmatic rocks in southern Tibet implies that the Neo-Tethys Ocean evolved continuously during the Late Jurassic and Early Cretaceous (Jiang et al., 2006; Meng et al., 2016). Paleogeographic

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reconstructions indicate that the southern margin of Asia was located at ~10°N during the Early Cretaceous. The Lhasa terrane did not undergo significant latitudinal movement during the Cretaceous (Sun et al., 2012; Tang et al., 2013; Cao et al., 2017a; Ma et al., 2022; Tong et al., 2022) because of the final collision of the Lhasa and Qiangtang terranes. In other words, most of the expansion of the Neo-Tethys Ocean occurred between the Late Triassic and the Early Cretaceous, at which time the Neo-Tethys Ocean reached its maximum width of at least 4185 km (Fig. 9). The difference in paleolatitude of the Lhasa terrane between the Late Triassic and the Early Cretaceous suggests an average northward drift rate of ~5 cm/yr for the terrane (Li et al., 2016), which is consistent with the calculated plate-tectonic motions and subduction rates (Van Der Meer et al., 2014). The Neo-Tethys Ocean then shrank after the Indian plate drifted away from Gondwana during the Early Cretaceous and was completely closed following the India-Asia collision during the Paleogene.

CONCLUSIONS

We acquired new paleomagnetic data from volcanic rocks dated to ca. 125 Ma within the eastern segment of the Lhasa terrane. Petrographic investigations together with fold, reversal, and PSV tests confirm that the rocks recorded the primary remanent direction, and they yielded a paleomagnetic pole of 60.9°N, 227.2°E ($dp/dm = 2.4°/4.6°$), indicating an Early Cretaceous paleolatitude of 9.2 ± 2.4°N for the eastern segment of the Lhasa terrane. We also compiled reliable paleomagnetic data from previous studies. The combined data indicate that the Lhasa terrane was oriented WNW-ESE as a whole during the Early Cretaceous, and that the eastern-middle segments may have been oriented nearly E-W at this time. Compared with the APWP of Asia, we find that Asia has accommodated 2050 ± 230 km of N-S crustal shortening along 96°E longitude since the Early Cretaceous. Furthermore, the paleolatitudinal difference between the eastern segment of the Lhasa terrane and the western segment of the Tethyan Himalaya terrane implies that the Neo-Tethys Ocean had a minimum N-S width of 4185 ± 300 km at ca. 125 Ma.

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